Weerahannadige Milinda Shayamal Fernando

Contact Information

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Objective

General purposes

Research Interests

Keywords: Numerical Relativity, Binary Compact Mergers, Space Filling Curves (SFCs), Parallel Computing, Partial Differential Equations (PDEs), Hierarchical Semi-Separable(HSS) Matrices, Numerical Linear Algebra, Scientific Computing

My research mainly focuses on developing large scalable parallel algorithms for simulating, 1). Massively parallel intermediate mass ratio inspirals (IMRIs) 2). Galerkin (both continuous and discontinuous) methods on parallel adaptive octrees, 3). Hierarchical Semi-Separable matrix factorizations 4). Octree based adaptive mesh generation, $2:1$ balancing constraint on octrees.

Education

University of Utah, Utah, UT

Ph.D., Scientific Computing, *Expected:* 2020

- Current:
  - ∗ **DENDRO-GR**: *Massively parallel simulations of binary black hole intermediate-mass-ratio inspirals*
  - ∗ **DENDRO**: Mesh-free finite element computations on adaptive octrees.
  - ∗ **SFC based octree partitioning**:*Fast scalable parallel sorting algorithms for Hilbert SFCs*
- Advisors: Hari Sundar, Ph.D

University of Moratuwa, Moratuwa, Sri Lanka

B.Sc., Computer Science and Engineering, April 2015

- *Final Year Research Project: Auto-Tuning the Java Virtual Machine using OpenTuner Framework*

Research Experience

Research Assistant — Augest 2015 to present
School of Computing,
University of Utah
Supervisor: Hari Sundar, Ph.D

Junior Consultant — April 2015 to August 2015
Department of Computer Science and Engineering,
University of Moratuwa
Supervisor: Sanath Jayasena, Ph.D

Refereed Paper Publications

1. Milinda Fernando, David Neilsen, Hyun Lim, Eric Hirschmann, Hari Sundar, "Massively Parallel Simulations of Binary Black Hole Intermediate-Mass-Ratio Inspirals" (arXiv:1807.06128v1) (Submited to SIAM SISC journal)
2. Milinda Fernando, Dmitry Duplyakin, and Hari Sundar. 2017. "Machine and Application Aware Partitioning for Adaptive Mesh Refinement Applications". In Proceedings of the 26th International Symposium on High-Performance Parallel and Distributed Computing (HPDC '17). ACM, New York, NY, USA, 231-242. DOI: https://doi.org/10.1145/3078597.3078610

3. Isuru Fernando, Sanath Jayasena, Milinda Fernando, Hari Sundar, "A Scalable Hierarchical Semi-Separable Library for Heterogeneous Clusters" 2017 46^{th} International Conference on Parallel Processing (ICPP), Bristol, UK, 2017

Refereed Poster Publications

1. Fernando, M., Duplyakin, D., Sundar,H. Energy and Communication Efficient Partitioning for Large-Scale Finite Element Computations,Poster in International Conference for High Performance Computing, Networking, Storage and Analysis (SC16)
2. Fernando, M. Sundar, H. Hilbert Curve based Flexible Dynamic Partitioning Scheme for Adaptive Scientific Computations, Poster (ACM SRC) in International Conference for High Performance Computing, Networking, Storage and Analysis (SC15)
3. Fernando, M. Rusira, T. Perera,C. Phillips, C. Auto-Tuning the Hotspot Java Virtual Machine Poster in International Symposium on Code Generation and Optimization(CGO15)

Refereed Workshop Publications

1. Jayasena, S.; Fernando, M.; Rusira, T.; Perera, C.; Philips, C., "Auto-Tuning the Java Virtual Machine," in Parallel and Distributed Processing Symposium Workshop (IPDPSW), 2015 IEEE International , vol., no., pp.1261-1270, 25-29 May 2015

Awards

Academic Awards

- Best Poster Award (Undergraduate Category)in CGO 2015, ACM Student Research Competition
- First Class Honours in B.Sc in Computer Science and Engineering
- NSF travel grant award (HPDC'17)

Student Awards — University of Utah, Graduate School

- School of Computing Fellowship 2015–2016

Presentations

- 2018 North American Einstein Toolkit Workshop (on DENDRO-GR) June 18-20

Teaching Experience

Teaching Assistant Spring 2017
CS 6230 - Parallel & High Performance Computing
with Hari Sundar
School of Computing,
University of Utah

Teaching Assistant Fall 2015
CS 5965/6965 - Big Data Computer Systems
with Hari Sundar
School of Computing,
University of Utah

References

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University of Utah

David Neilsen
Professor — Phone: : 801-422-6078
Department of Physics and Astronomy — E-mail: david.neilsen @byu.edu
Brigham Young University